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Reserved IP addresses

In the Internet addressing architecture, the Internet Engineering Task Force (IETF) and the Internet Assigned Numbers Authority (IANA) have reserved various Internet Protocol (IP) addresses for special purposes.^[1]

IPv4

IPv4 designates special usage or applications for various addresses or address blocks:^{[1][2]}

Special address blocks

Address block	Address range	Number of addresses	Scope	Description
0.0.0.0/8	0.0.0.0–0.255.255.255	16 777 216	Software	Current (local, "this") network ^[1]
10.0.0.0/8	10.0.0.0–10.255.255.255	16 777 216	Private network	Used for local communications within a private network ^[3]
100.64.0.0/10	100.64.0.0–100.127.255.255	4 194 304	Private network	Shared address space ^[4] for communications between a service provider and its subscribers when using a carrier-grade NAT
127.0.0.0/8	127.0.0.0–127.255.255.255	16 777 216	Host	Used for <u>loopback addresses</u> to the local host ^[1]
169.254.0.0/16	169.254.0.0–169.254.255.255	65 536	Subnet	Used for link-local addresses ^[5] between two hosts on a single link when no IP address is otherwise specified, such as would have normally been retrieved from a <u>DHCP</u> server
172.16.0.0/12	172.16.0.0–172.31.255.255	1 048 576	Private network	Used for local communications within a private network ^[3]
192.0.0.0/24	192.0.0.0–192.0.0.255	256	Private network	IETF Protocol Assignments, <u>DS-Lite</u> (/29) ^[1]
192.0.2.0/24	192.0.2.0–192.0.2.255	256	Documentation	Assigned as TEST-NET-1, documentation and examples ^[6]
192.88.99.0/24	192.88.99.0–192.88.99.255	256	Internet	Reserved. ^[7] Formerly used for IPv6 to IPv4 relay ^[8] (included IPv6 address block 2002::/16).
192.168.0.0/16	192.168.0.0–192.168.255.255	65 536	Private network	Used for local communications within a private network ^[3]
198.18.0.0/15	198.18.0.0–198.19.255.255	131 072	Private network	Used for benchmark testing of inter-network communications between two separate subnets ^[9]
198.51.100.0/24	198.51.100.0–198.51.100.255	256	Documentation	Assigned as TEST-NET-2, documentation and examples ^[6]
203.0.113.0/24	203.0.113.0–203.0.113.255	256	Documentation	Assigned as TEST-NET-3, documentation and examples ^[6]
224.0.0.0/4	224.0.0.0–239.255.255.255	268 435 456	Internet	In use for multicast ^[10] (former Class D network)

Address block	Address range	Number of addresses	Scope	Description
233.252.0.0/24	233.252.0.0–233.252.0.255	256	Documentation	Assigned as MCAST-TEST-NET, documentation and examples (This is part of the above multicast space.) ^[10] ^[11]
240.0.0.0/4	240.0.0.0–255.255.255.254	268 435 455	Internet	Reserved for future use ^[12] (former Class E network)
255.255.255.255/32	255.255.255.255	1	Subnet	Reserved for the "limited broadcast" destination address ^[1]

IPv6

IPv6 assigns special uses or applications for various IP addresses:^[1]

Special address blocks

Address block (CIDR)	First address	Last address	Number of addresses	Usage	Purpose
::/128	::	::	1	Software	Unspecified address
::1/128	::1	::1	1	Host	<u>Loopback address</u> —a virtual interface that loops all traffic back to itself, the <i>local host</i>
::ffff:0:0/96	::ffff:0:0:0:0 ::ffff:0:0	::ffff:255.255.255.255 ::ffff:ffff:ffff	2 ³²	Software	IPv4-mapped addresses
::ffff:0:0:0/96	::ffff:0:0:0:0:0 ::ffff:0:0:0	::ffff:0:255.255.255.255 ::ffff:0:ffff:ffff	2 ³²	Software	IPv4-translated addresses
64:ff9b::/96	64:ff9b::0:0:0:0 64:ff9b::0:0	64:ff9b::255.255.255.255 64:ff9b::ffff:ffff	2 ³²	The global Internet	IPv4/IPv6 translation ^[13]
64:ff9b:1::/48	64:ff9b:1::	64:ff9b:1:ffff:ffff:ffff:ffff:ffff	2 ⁸⁰ , with 2 ⁴⁸ for each IPv4	Private internets	IPv4/IPv6 translation ^[14]
100::/64	100::	100::ffff:ffff:ffff:ffff	2 ⁶⁴	Routing	Discard prefix ^[15]
2001::/32	2001::	2001:0:ffff:ffff:ffff:ffff:ffff:ffff	2 ⁹⁶	The global Internet	<u>Teredo tunneling</u> ^[16]
2001:20::/28	2001:20::	2001:2f:ffff:ffff:ffff:ffff:ffff:ffff	2 ¹⁰⁰	Software	<u>ORCHIDv2</u> ^[17]
2001:db8::/32	2001:db8::	2001:db8:ffff:ffff:ffff:ffff:ffff:ffff	2 ⁹⁶	Documentation	Addresses used in documentation and example source code ^[18]
2002::/16	2002::	2002:ffff:ffff:ffff:ffff:ffff:ffff:ffff	2 ¹¹²	The global Internet	The 6to4 addressing scheme (deprecated) ^[7]
3fff::/20	3fff::	3fff:fff:ffff:ffff:ffff:ffff:ffff:ffff	2 ¹⁰⁸	Documentation	Addresses used in documentation and example source code ^[19]
5f00::/16	5f00::	5f00:ffff:ffff:ffff:ffff:ffff:ffff:ffff	2 ¹¹²	Routing	<u>IPv6 Segment Routing (SRv6)</u> ^[20]
fc00::/7	fc00::	fdff:ffff:ffff:ffff:ffff:ffff:ffff:ffff	2 ¹²¹	Private internets	<u>Unique local address</u> ^[21]
fe80::/64 from fe80::/10	fe80::	fe80::ffff:ffff:ffff:ffff	2 ⁶⁴	Link	<u>Link-local address</u>
ff00::/8	ff00::	ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff	2 ¹²⁰	The global Internet	<u>Multicast address</u>